

Master Technical Architecture Specification

Complete End-to-End Architectural Reference & Implementation Guide for the RVCC Digital Procurement, Real-Time Bidding, and Enterprise Resource Planning Ecosystem.

5 Apps

TURBOREPO PACKAGES

65,453

LINES OF TYPESCRIPT

31 Models

PRISMA RELATIONAL DB

100% Pass

VITEST & CI TESTS

System Version: 2.0.0 (Production Release)

Stack: Next.js 16 • Turbopack • Cloudflare Edge • Prisma ORM

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1. Monorepo & Shared Architecture

The **RVCC Digital Ecosystem** is structured as an enterprise-grade Turborepo monorepo powered by PNPM workspaces. It isolates concerns across 5 full-stack applications and 3 shared internal packages while maximizing code reuse and type safety.

```
RVCC-MONOREPO/
├─ apps/
│  ├─ web/           # Port 3000 - Public Corporate & Vendor Onboarding Portal (Next.js 16)
│  ├─ admin/         # Port 3001 - Enterprise Super-Admin & Sourcing Governance ERP (Next.js 16)
│  ├─ vendor/         # Port 3002 - Vendor Portal & Real-Time Blind Bidding Cockpit (Next.js 16)
│  ├─ procurement/    # Port 3003 - Internal Department Requisition & Approval Engine (Next.js 16)
│  └─ api/            # Port 8787 - Edge Backend API, SSE Streaming Engine & Prisma ORM
└─ packages/
   ├─ schemas/        # @rvcc/schemas - Shared Zod validation contracts (Single Source of Truth)
   ├─ types/          # @rvcc/types - Centralized TypeScript interfaces & DB entity shapes
   └─ utils/          # @rvcc/utils - Common currency math, string sanitizers & formatters
```

Shared Packages Matrix

Package	Exports & Key Modules	Role & Architecture Benefit
@rvcc/schemas	auth.ts, procurement.ts, sourcing.ts, vendor-onboarding.ts, rbac.ts, audit.ts, enums.ts	Runtime schema validation using Zod. Guaranteed boundary validation on all API requests and frontend form submissions.
@rvcc/types	Database entity types, API response generics (ApiResponse<T>), Live Bidding contracts, Currency models.	Compile-time type safety preventing contract drift between backend edge workers and frontend clients.
@rvcc/utils	formatters.ts, currency.ts, sanitize.ts, rank.ts, cn.ts	High-performance utility functions for SAR multi-currency conversions, L1 ranking algorithms, and styling.

CI/CD & DevOps Automation

- GitHub Actions CI:** Automated test, lint, and build validation on every pull request.
- Multi-Cloud Deploy:** Cloudflare Workers for apps/api and Vercel for Next.js applications.
- Turborepo Cache:** Remote caching with dependency graph hashing.

Code Standards & Tooling

- ESLint 9 Flat Config:** Automated unused import removal and simple-import-sort ordering.
- Prettier + Tailwind Plugin:** Deterministic formatting and automatic class sorting.
- Vitest Test Suites:** 10/10 automated tests running across all apps.

2. Public Corporate & Vendor Onboarding Application

APPS/WEB (PORT 3000)

`apps/web` serves as RVCC's corporate platform, interactive civil/MEP project showcase, and prospective supplier qualification portal. Built with Next.js 16 (App Router) and WebGL graphics.

Key Features & Routing Architecture

Route Path	Rendering Strategy	Technical Functionality
<code>/</code> (Home)	Static Prerender (SSG)	Hero with 3D Skyscraper WebGL canvas, Divisions showcase, Projects marquee, CSR initiatives.
<code>/about</code>	Static Prerender (SSG)	Corporate journey timeline, Mission/Vision/Values, Leadership, ISO certifications grid.
<code>/services/[slug]</code>	Incremental Static (ISR)	Dynamic service pages (Artificial Grass, MEP, Civil, Landscaping) with pre-generated static params.
<code>/projects/[slug]</code>	Dynamic Server / SSR	High-resolution architectural photography, client metadata, engineering specifications.
<code>/documents</code>	Static + Dynamic Gate	Document showcase with secure PDF Flipbook Reader (<code>FlipbookReader.tsx</code>) and password unlock.
<code>/careers</code>	Dynamic SSR + SWR	Live job vacancies board with resume submission and automated multipart upload API.
<code>/enquire/[step]</code>	Client Wizard Train	10-step supplier qualification train with draft recovery, document upload, and email OTP verification.

10-Step Supplier Qualification Train Matrix

Step	Domain Area	Form Data & Business Validation
1 - 2	Company & Addresses	Trade Name, Commercial Registration (CR), VAT Number, Establishment Year, HQ and Branch GPS coordinates.
3 - 4	Classifications & Bank	Primary/Secondary industry classifications, Beneficiary Name, Bank Name, IBAN, SWIFT code, Currency.
5 - 6	Contacts & Questionnaire	Executive contacts, Procurement representatives, ISO compliance disclosures, safety track record.
7 - 8	Attachments & Review	CR Certificate, VAT Proof, Chamber of Commerce membership PDF, consolidated preview terminal.
9 - 10	OTP & Finalize	Cryptographic 6-digit email OTP verification, tracking reference generation (<code>REG-XXXXXX</code>).

Interactive 3D WebGL & Physics Cards
`SkyscraperCanvas.tsx` utilizes Three.js with custom lighting and camera mouse-parallax tracking. `Interactive3DCard.tsx` implements hardware-accelerated CSS 3D perspective transforms.

3. Admin ERP & Sourcing Governance Portal

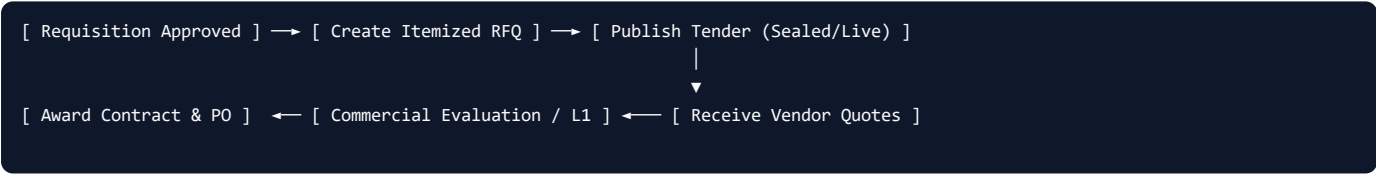
APPS/ADMIN (PORT 3001)

`apps/admin` is the centralized ERP and governance platform for procurement executives, sourcing officers, and auditors. It provides strict Role-Based Access Control, tender creation, live auction monitoring, and supplier verification.

Role-Based Access Control (RBAC) Hierarchy

Role Name	Granular Permissions Assigned	Administrative Scope
SUPER_ADMIN	All 12 Permissions (<code>MANAGE_STAFF</code> , <code>MANAGE_REQUIREMENTS</code> , <code>AWARD_REQUIREMENTS</code> , etc.)	Full governance, security policies, system configuration, user provisioning, and audit log inspection.
PROCUREMENT_MANAGER	<code>VIEW PROCUREMENT</code> , <code>MANAGE PROCUREMENT</code> , <code>VIEW REQUIREMENTS</code> , <code>AWARD_REQUIREMENTS</code>	Requisition approvals, tender publishing, quote evaluation, and contract awarding.
SOURCING_OFFICER	<code>VIEW REQUIREMENTS</code> , <code>MANAGE_REQUIREMENTS</code> , <code>VIEW VENDORS</code> , <code>MANAGE VENDORS</code>	BOQ creation, vendor invite matching, live bidding monitoring, and registration document review.
AUDITOR / COMPLIANCE	<code>VIEW REQUIREMENTS</code> , <code>VIEW VENDORS</code> , <code>VIEW_AUDIT_LOGS</code> , <code>VIEW_FINANCIALS</code>	Read-only inspection across all bidding histories, financial quotes, and system mutation logs.
CONTENT_MANAGER	<code>VIEW_CONTENT</code> , <code>MANAGE_CONTENT</code>	CMS management for corporate website services, projects, careers, and media gallery.

Sourcing & RFQ Lifecycle



Live Market Monitoring Cockpit

- **Real-Time SSE Feed:** Sub-second updates as vendors submit price revisions.
- **L1 Ranking Engine:** Instant recalculation of lowest bids normalized to SAR.
- **Anonymized View:** Toggle between blind view (for bias-free evaluation) and unmasked view.

Supplier Document Verification

- **Split-Screen Inspector:** Review CR licenses, tax filings, and bank documents.
- **Status Transitions:** Approve, request amendments, reject, or suspend accounts.
- **Auto-Provisioning:** Instantly provisions vendor accounts on approval.

4. Vendor Portal & Blind-Bidding Cockpit

APPS/VENDOR (PORT 3002)

`apps/vendor` provides registered suppliers with an external workspace to discover tender invitations, enter live blind auctions, submit multi-currency commercial bids, and manage compliance records.

Blind-Bidding Mechanics & Anonymization

To eliminate market collusion and guarantee fair competition, the Live Bidding Cockpit (`/requirements/[id]`) implements cryptographic participant anonymization:

VENDOR LIVE BIDDING COCKPIT

YOUR CURRENT RANK: #2
Quote: 45,000 SAR

CURRENT L1 PRICE (LOWEST)
42,500 SAR

TIME REMAIN
02h : 14m

RANKING TABLE (ANONYMIZED)

- Rank #1: 42,500 SAR (Lowest Bid - Market Leader)
- Rank #2: 45,000 SAR (YOU - Delta: +2,500 SAR / +5.8%)
- Rank #3: 48,000 SAR (Competitor B)
- Rank #4: 52,000 SAR (Competitor C)

INSTANT PRICE REVISION CONTROLLER
[Enter Unit Price] [Match L1 Target (42,500 SAR)] [SUBMIT BID]

Multi-Currency Normalization Engine (SAR Baseline)

Currency	Sync Mechanism	Conversion Formula	Display & Ranking
SAR (Base)	1.0000 Baseline	<code>amountSar = amount</code>	Official currency for all L1 ranking comparisons.
USD	Daily 24h Cron Sync	<code>amountSar = amount * rateUsd</code>	Displays vendor quote in USD with live normalized SAR equivalence.
EUR / GBP	Daily 24h Cron Sync	<code>amountSar = amount * rateEur</code>	Real-time automatic conversion preventing currency volatility bias.
AED / QAR	Daily 24h Cron Sync	<code>amountSar = amount * rateGcc</code>	Supports regional GCC currency quoting natively.

Resilient Streaming (use-vendor-live-bidding.ts)

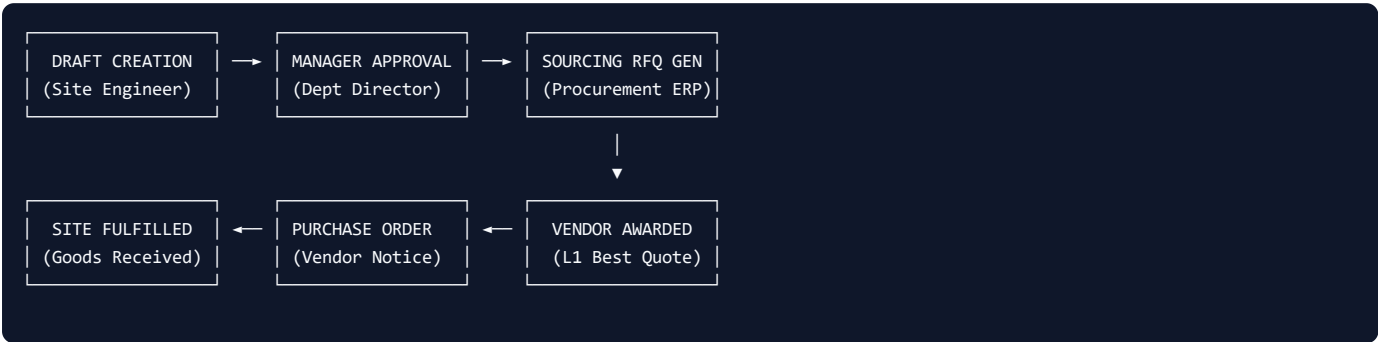
Connects via Server-Sent Events (SSE) for zero-latency bid updates. If network degradation occurs, it automatically falls back to a 5-second polling loop and seamlessly recovers the SSE connection when connectivity stabilizes.

5. Internal Procurement & Requisition App

APPS/PROCUREMENT (PORT 3003)

`apps/procurement` is the dedicated internal purchase requisition platform designed for department managers, site engineers, and project controllers to initiate and track material/service requests.

Purchase Requisition Lifecycle



Requisition Data Model & Attributes

Attribute	Data Type	Description & Business Logic
<code>requisitionNumber</code>	String (Unique)	Sequential reference code (e.g. <code>PR-2026-0042</code>).
<code>department</code>	Enum (Department)	Civil, MEP, Landscaping, Infrastructure, Architecture, Logistics.
<code>priorityLevel</code>	Enum (Priority)	<code>LOW</code> , <code>STANDARD</code> , <code>URGENT</code> , <code>EMERGENCY</code> (controls approval SLA).
<code>estimatedBudget</code>	Decimal (SAR)	Estimated spend checked against departmental budget allocations.
<code>items</code>	Array<BOQItem>	Item name, technical specifications, quantity, unit of measure, estimated rate.
<code>attachments</code>	Array<File>	BOQ spreadsheets, technical datasheets, site engineering drawings.

Mobile & On-Site Engineering UX

- **PWA Offline Support:** Create requisition drafts on construction sites without active internet.
- **Quick-Action Modal:** Rapid 3-click requisition submission with voice-to-text note capture.
- **Status KPI Badges:** Real-time colored visual cues (`DRAFT`, `APPROVED`, `SOURCING`, `ORDERED`).

Approval Matrix & Budget Control

- **Threshold Routing:** Requisitions over defined SAR thresholds require Finance Director sign-off.
- **Rejection Feedback:** Comprehensive justification notes returned to the requesting engineer.
- **Audit Logging:** All approval and rejection timestamps immutably recorded.

6. Edge API & Database Architecture

`apps/api` is the high-throughput Edge Backend API. Powered by Cloudflare Workers / Node.js Edge Runtime and Prisma ORM v5.22 with the `@prisma/adapters-pg` connection driver, it handles sub-millisecond query execution, real-time SSE streaming, and secure authentication.

Complete 31-Model Relational Database Architecture

Domain Group	Prisma Entity Models	Architectural Function
1. Staff & RBAC	<code>AdminUser</code> , <code>Role</code> , <code>Permission</code> , <code>RolePermission</code> , <code>AdminSession</code> , <code>AdminOtp</code> , <code>AdminLoginHistory</code>	Granular role-permission mappings, secure session tokens, password reset OTP challenges, and chronological login history.
2. Vendor & Auth	<code>VendorUser</code> , <code>VendorSession</code> , <code>VendorOtp</code> , <code>VendorLoginHistory</code> , <code>SupplierRegistration</code>	Vendor authentication, passwordless OTP verification, session lifecycle, and registration dossiers.
3. Supplier Dossier	<code>CompanyProfile</code> , <code>SupplierContact</code> , <code>SupplierAddress</code> , <code>BusinessClassification</code> , <code>BankAccount</code> , <code>QuestionnaireAnswer</code> , <code>RegistrationAttachment</code>	Complete 7-part supplier qualification records including legal, banking, contact, and compliance documents.
4. Sourcing & Bidding	<code>Requirement</code> , <code>RequirementInvite</code> , <code>Quote</code> , <code>ExchangeRate</code> , <code>Industry</code>	Tender requirements, itemized BOQ, targeted supplier invites, multi-currency quotes, and daily FX rates.
5. Procurement PR	<code>PurchaseRequest</code> , <code>PurchaseRequestItem</code> , <code>PurchaseRequestAttachment</code>	Purchase requisitions, line item breakdowns, approval history, and direct links to sourcing tenders.
6. Audit & Content	<code>Notification</code> , <code>AuditLog</code> , <code>JobPosting</code> , <code>JobApplication</code>	Real-time user notifications, immutable audit log ledger with JSON state diffs, careers CMS and candidate applications.

Prisma Soft-Delete Extension

```
// apps/api/src/db.ts - Soft-Delete Safeguard
export const prisma = new PrismaClient().$extends({
  query: {
    $allModels: {
      async delete({ model, args }) {
        return (prisma[model] as any).update({ ...args, data: { deletedAt: new Date() } });
      },
      async findMany({ model, args }) {
        args.where = { deletedAt: null, ...args.where };
        return (prisma[model] as any).findMany(args);
      }
    }
  }
});
```

Real-Time SSE Streaming Bus & Security Governance

- **Server-Sent Events (SSE):** Sub-second broadcasting of revised bids with automated delta rankings. Dual payloads prevent competitor identity leaks to vendor clients while giving administrators full visibility.
- **Automated Daily FX Sync:** Background cron service queries official exchange rates for USD, EUR, GBP, AED, QAR against SAR and caches them in memory.
- **Zero-Trust Session Security:** HttpOnly, Secure, SameSite=Lax JWT cookies preventing XSS token exfiltration. Timing-safe OTP hash comparisons prevent timing attack exploits.

- **Immutable Audit Logging:** Every state change across tenders, quotes, approvals, and user accounts is logged with before/after JSON diffs, user ID, IP address, and microsecond timestamps.